

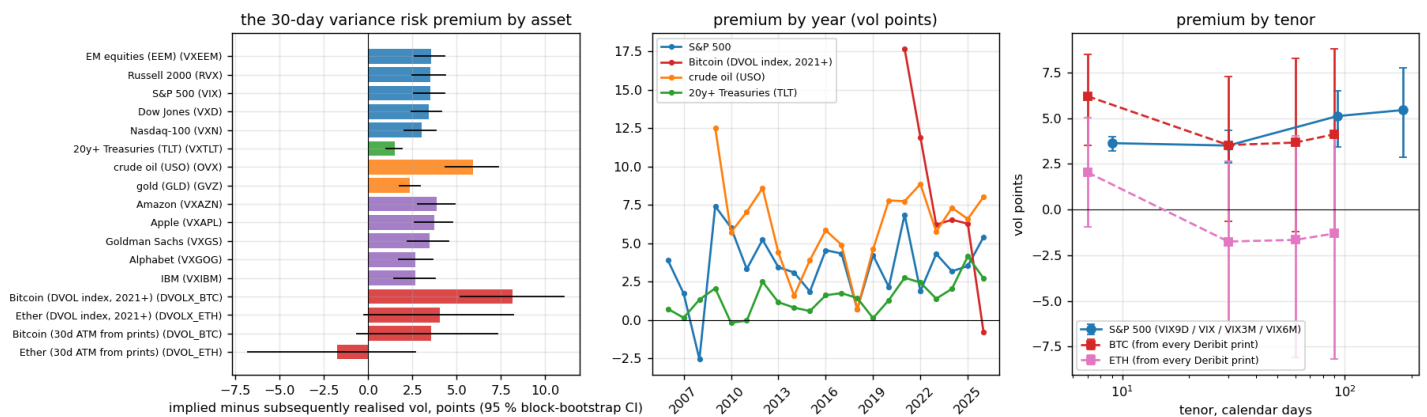
Volatility Dynamics Research and Systematic Vol Strategies

Question. Which volatility dynamics are stable enough to trade? On complete free histories: the gap between implied and subsequently realised variance by asset and tenor, how well realised variance can be forecast and whether implied vol adds to the forecast, the size and persistence of the earnings and FOMC premia, the VIX term structure and the roll of its futures, the calendar of a 24/7 market, and what each of these leaves after bid-ask, hedging costs and margin when turned into a systematic strategy.

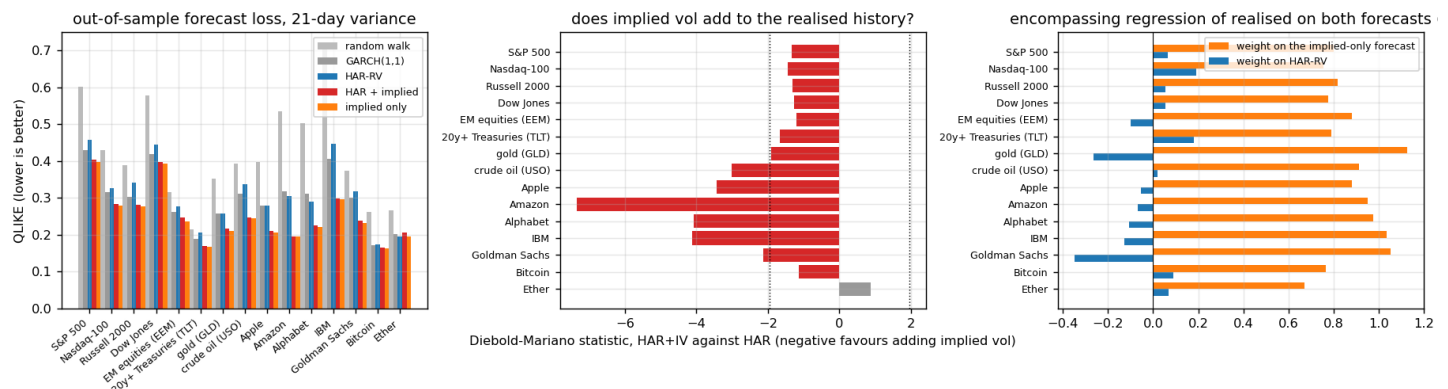
Data. Cboe daily histories of nineteen volatility indices (VIX family, VVIX, SKEW, VXN, RVX, VXD, GVZ, OVX, VXTLT, VXEEM and five single-name indices) and every VX futures contract since 2013; every Deribit BTC and ETH option print since the first trade (Nov 2016 / Mar 2019) with the exchange's implied vol, plus 5-minute perpetual candles and DVOL; DoltHub end-of-day option chains and earnings dates for SPY and fifty large caps; Yahoo daily bars; SOFR from the New York Fed; the FOMC calendar. The surface engine (Black, implied vol, SSVI slices, forwards from parity) is Project H's.

Method. Realised variance from OHLC estimators and 5-minute returns with the Barndorff-Nielsen-Shephard jump test. HAR-RV, HAR with implied variance, GARCH(1,1) and a random walk fitted walk-forward and scored by QLIKE with Diebold-Mariano tests. The premium as implied variance minus forward realised variance with circular block-bootstrap intervals, by asset, tenor, year and regime. Skew stickiness from daily SSVI fits. Earnings: the implied move stripped of the diffusion from two expiries against the realised move, and the straddle seller's P/L after the quoted spread. FOMC: the path of the short-dated indices and realised against implied on decision days. The VX curve, roll-down and the short-front roll net of bid-ask. Strategies in an engine that marks Black-Scholes daily, hedges delta daily at a stated cost, settles at intrinsic and quotes return per unit of a scenario margin; parameters chosen on the first half and reported on the second.

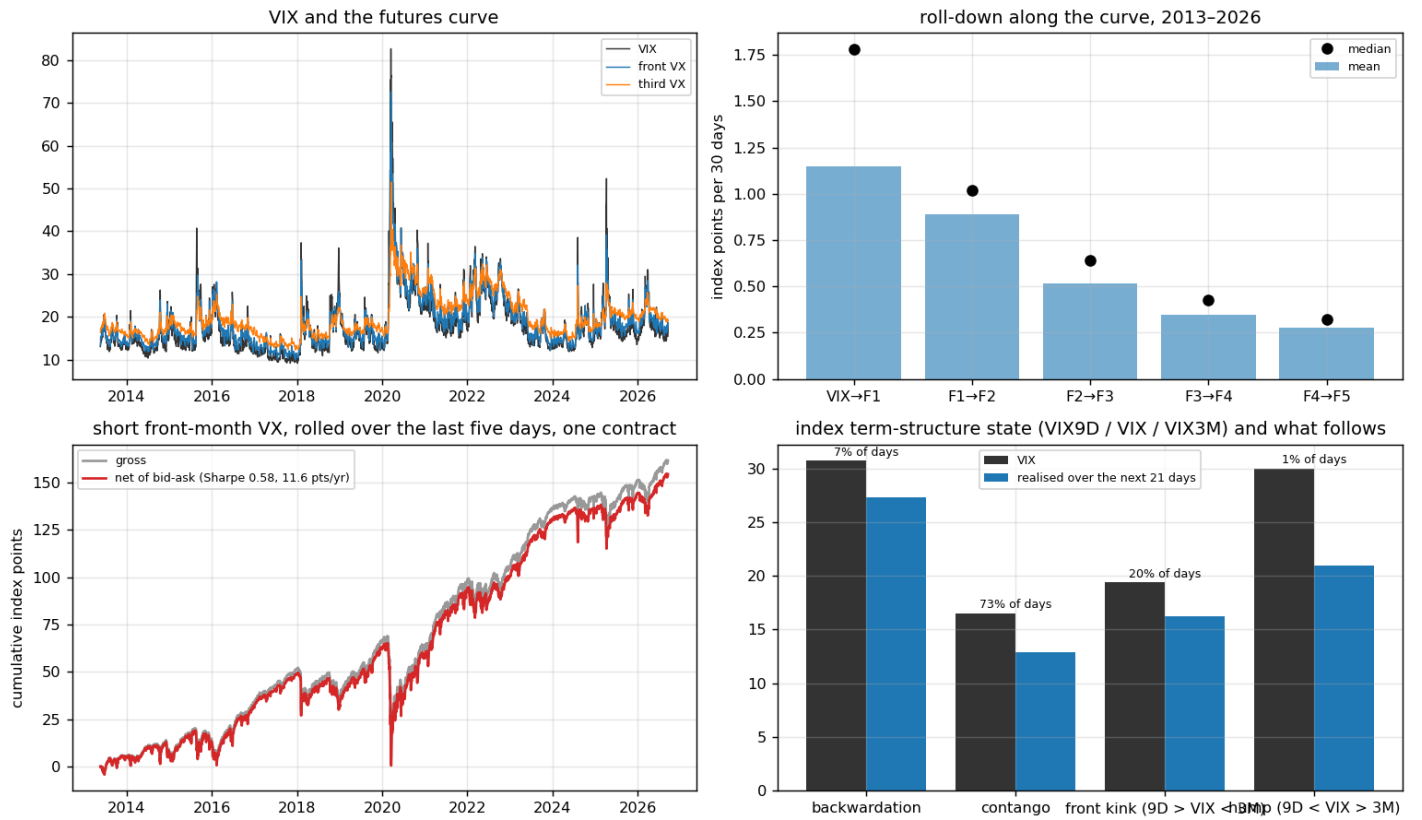
Caveats. The DoltHub chains are sparse (about three expiries and twenty strikes a day, weekly before 2023), so the SPY strategies are marked to model between chain dates and settled on the real close. Crypto option spreads are an assumption stated in the code. The OTC vol markets have no free history. Results are stated with the years they cover; a single regime is not allowed to carry an average without saying so.



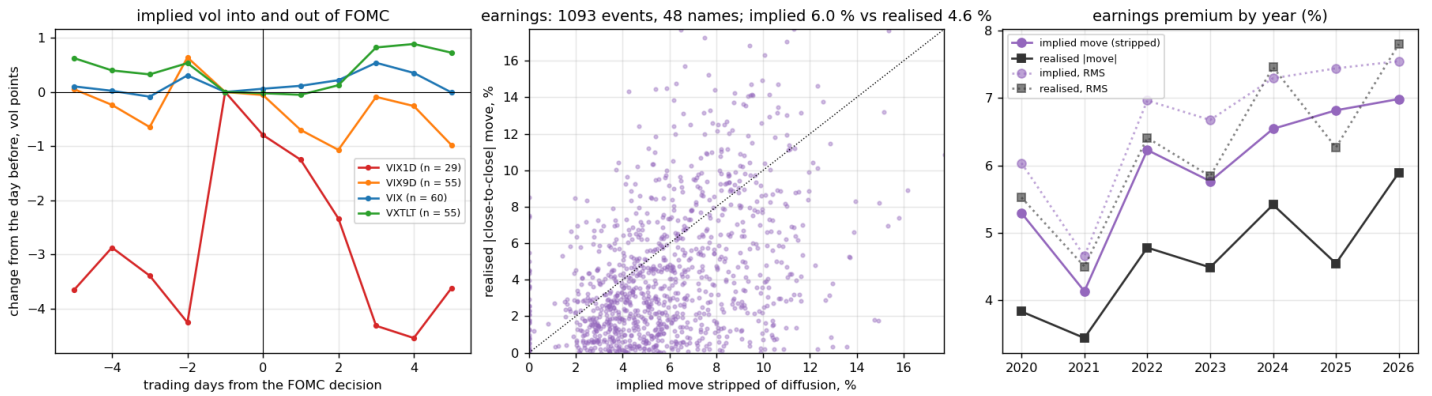
The variance risk premium: by asset with block-bootstrap intervals, by year, and by tenor for the S&P and for BTC and ETH.



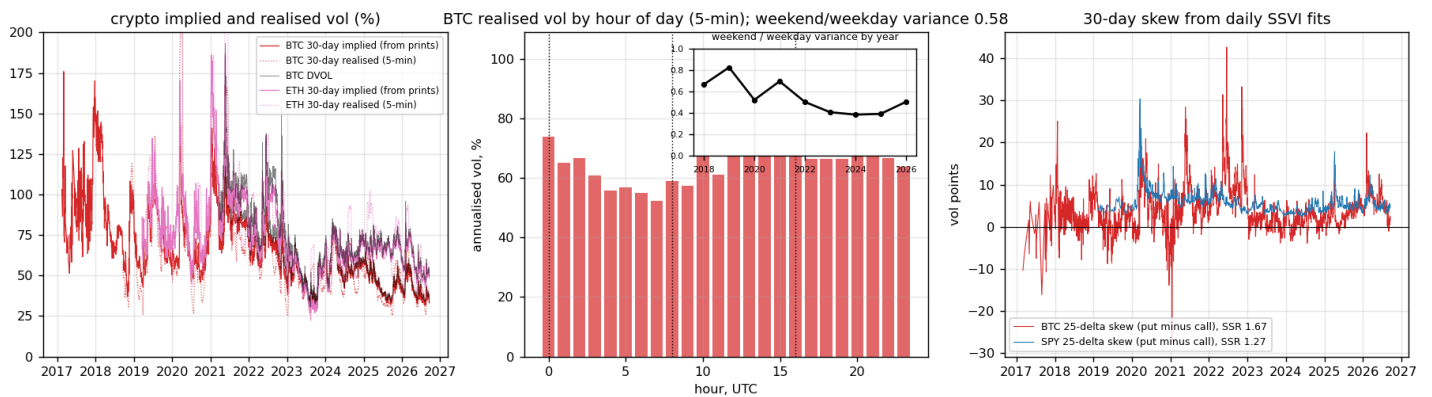
Forecasting realised variance: QLIKE by model and asset, the Diebold-Mariano statistic of adding implied vol to HAR, and the encompassing weights.



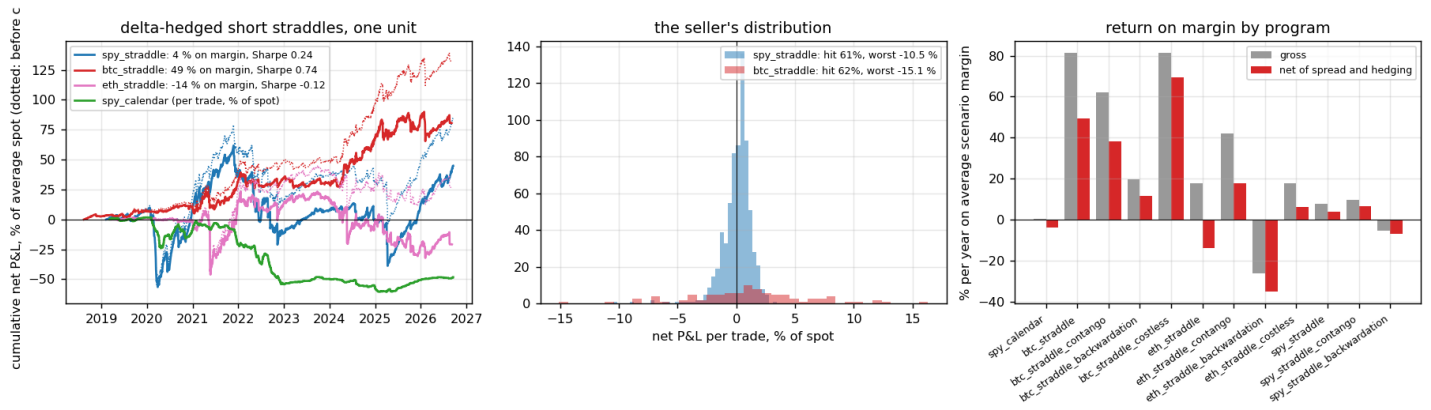
The VIX complex: index and futures, roll-down along the curve, the short-front roll net of bid-ask, and what follows each term-structure state.



Events: implied vol into and out of FOMC decisions; earnings implied against realised moves and the premium by year.



Crypto: print-implied against realised vol and DVOL, realised vol by hour of day with the weekend ratio, and the 30-day skew from daily SSVI fits.



Strategies: cumulative net P&L of the delta-hedged straddle programs, the seller's per-trade distribution, and return on margin gross against net.

Generated from results/*.json by scripts/summarize.py.

Data built

- BTC: 12,218,677 near-the-money prints into 3,406 days of implied term structure (2017-02-14 to 2026-09-18); 2,956 days of 5-minute realised variance, annualised 67 % (close-to-close 64 %); BNS jump days 32% at 5 %, 26% at 1 %, jump variation 6% of RV; 30-day print-implied against DVOL: corr 0.985, mean gap -4.03 pts on 2,005 days; SSVI slices 3,459 days, median RMSE 1.77 vol points, butterfly-free 100 %.
- ETH: 7,153,447 near-the-money prints into 2,738 days of implied term structure (2019-03-21 to 2026-09-18); 2,744 days of 5-minute realised variance, annualised 85 % (close-to-close 83 %); BNS jump days 27% at 5 %, 22% at 1 %, jump variation 4% of RV; 30-day print-implied against DVOL: corr 0.986, mean gap -4.94 pts on 2,005 days; SSVI slices 2,739 days, median RMSE 1.57 vol points, butterfly-free 100 %.
- SPY: 1,267 daily SSVI slices (2019-02-09 to 2026-09-17), median RMSE 0.98 vol points, butterfly-free 100 %, mean 25-delta skew 5.8 points.
- US daily estimators for 13 underlyings, 65,403 rows; mean ratio of each estimator to close-to-close: park 0.579, gk 0.567, rs 0.571, yz21 0.928.

The variance risk premium

30-day premium by asset: implied (the Cboe index or, for crypto, the print-implied 30-day vol) against the vol realised over the following 21 trading days (30 calendar days for crypto). Intervals are circular block bootstraps with the horizon as block length. Slope: realised variance regressed on implied variance (Newey-West s.e.).

											implied ind
DVOLX_BTC	Bitcoin (DVOL index, crypto)	2021	2026	1,974	63.1	55.4	8.16	[5.20, 11.05]	71 %	0.51 (0.08)	
OVX	crude oil (USO)	commodity	2009	2026	4,248	41.7	36.7	5.91	[4.35, 7.33]	79 %	0.42 (0.16)
DVOLX_ETH	Ether (DVOL index, crypto)	2021	2026	1,974	77.7	75.3	4.03	[-0.25, 8.20]	63 %	0.63 (0.13)	
VXAZN	Amazon	single name	2011	2026	3,919	35.4	32.9	3.85	[2.81, 4.91]	74 %	0.87 (0.06)
VXAPL	Apple	single name	2011	2026	3,919	30.7	28.3	3.72	[2.63, 4.74]	74 %	0.76 (0.10)
VXEEM	EM equities (EEM)	equity index	2011	2026	3,873	24.1	21.5	3.57	[2.62, 4.30]	81 %	0.64 (0.06)
DVOL_BTC	Bitcoin (30d ATM from crypto)	2018	2026	2,926	64.7	64.0	3.53	[-0.63, 7.30]	66 %	0.48 (0.08)	
RVX	Russell 2000	equity index	2009	2026	4,249	24.9	22.7	3.53	[2.47, 4.37]	82 %	0.89 (0.15)
VIX	S&P 500	equity index	2006	2026	5,007	21.5	19.6	3.51	[2.58, 4.33]	83 %	0.90 (0.12)
VXGS	Goldman Sachs	single name	2011	2026	3,919	31.2	29.0	3.46	[2.21, 4.54]	77 %	0.73 (0.11)
VXD	Dow Jones	equity index	2009	2026	4,249	18.4	16.5	3.41	[2.46, 4.15]	85 %	0.99 (0.31)
VXN	Nasdaq-100	equity index	2009	2026	4,256	22.6	20.8	3.00	[2.06, 3.80]	79 %	0.83 (0.13)
VXGOG	Alphabet	single name	2011	2026	3,919	29.4	27.9	2.67	[1.70, 3.64]	71 %	0.79 (0.08)
VXIBM	IBM	single name	2011	2026	3,919	26.1	25.7	2.64	[1.45, 3.76]	74 %	1.19 (0.27)
GVZ	gold (GLD)	commodity	2009	2026	4,248	18.6	16.9	2.36	[1.76, 2.91]	79 %	0.72 (0.11)
VXTLT	20y+ Treasuries (TLT)	2006	2026	4,997	16.1	14.9	1.48	[1.03, 1.89]	72 %	0.70 (0.08)	
DVOL_ETH	Ether (30d ATM from crypto)	2019	2026	2,707	77.9	83.0	-1.75	[-6.79, 2.65]	55 %	0.65 (0.12)	

S&P 500 premium by tenor (VIX family):

									index
VIX9D	9	3,941	19.3	17.2	3.64	[3.21, 4.01]	82 %		
VIX	30	5,007	21.5	19.6	3.51	[2.58, 4.33]	83 %		
VIX3M	93	4,209	21.3	17.2	5.12	[3.43, 6.50]	86 %		
VIX6M	184	4,578	23.9	19.9	5.45	[2.88, 7.76]	84 %		

BTC premium by tenor (print-implied term structure; the last row uses 5-minute realised variance as the realised leg):

							tenor
7	2,949	64.0	63.8	6.21	[3.52, 8.51]	69 %	
30	2,926	64.7	64.0	3.53	[-0.63, 7.30]	66 %	
60	2,896	66.0	64.1	3.67	[-1.23, 8.27]	67 %	
90	2,866	67.1	64.4	4.12	[-1.32, 8.80]	68 %	
30_rv5min	2,926	64.7	67.4	0.17	[-3.78, 3.93]	61 %	
30_dvol	1,974	63.1	55.4	8.16	[5.20, 11.05]	71 %	
30_atm_since_2021	1,974	58.5	55.4	4.09	[1.14, 6.92]	67 %	

ETH premium by tenor (print-implied term structure; the last row uses 5-minute realised variance as the realised leg):

						tenor
7	2,730	77.2	82.9	2.04	[-0.97, 5.04]	61 %

ProjectI - volatility dynamics research and systematic volatility strategies

30	2,707	77.9	83.0	-1.75	[-6.79, 2.65]	55 %
60	2,677	79.3	83.1	-1.66	[-8.09, 4.04]	55 %
90	2,647	80.3	83.1	-1.31	[-8.20, 4.31]	54 %
30_rv5min	2,707	77.9	85.3	-4.26	[-9.16, 0.05]	52 %
30_dvol	1,974	77.7	75.3	4.03	[-0.25, 8.20]	63 %
30_atm_since_2021	1,974	72.1	75.3	-0.96	[-5.27, 3.24]	56 %

S&P premium by regime (VIX terciles, then by the sign of VIX3M minus VIX on the day):

							regime
VIX low	1,669	12.9	2.25	20	[-32, 55]	82 %	
VIX mid	1,669	17.5	3.52	70	[24, 110]	83 %	
VIX high	1,669	28.6	4.75	141	[-42, 289]	84 %	
contango (long above short)	9,927	-	3.76	94	[53, 125]	85 %	
backwardation	325	-	4.38	63	[-535, 480]	83 %	

Skew dynamics from daily SSVI fits of the expiry nearest 30 days:

								surface
BTC	3,139	3.8	1.67 (0.08)	0.13	2.22	2.28	3.57 / 3.74	closer to sticky str
SPY	1,267	5.8	1.27 (0.03)	0.67	0.72	1.23	1.25 / 2.84	closer to sticky str

Forecasting realised variance

Walk-forward, refit every 21 days on the trailing window, horizon 21 trading days (30 for crypto). QLIKE and the RMSE of the forecast vol (daily units x 1e4); DM: Diebold-Mariano of HAR+IV against HAR on QLIKE (negative favours adding implied vol); encompassing weights from a log regression of realised on the implied-only and HAR forecasts.

										asset
S&P 500	3,988	0.60	0.43	0.46	0.40	0.40	-1.3 (0.184)	0.80 / 0.06	49.1 vs 51.6	
Nasdaq-100	3,988	0.43	0.32	0.33	0.28	0.28	-1.5 (0.146)	0.76 / 0.19	52.2 vs 53.2	
Russell 2000	3,983	0.39	0.30	0.34	0.28	0.28	-1.3 (0.193)	0.82 / 0.05	54.7 vs 58.7	
Dow Jones	3,985	0.58	0.42	0.44	0.40	0.39	-1.3 (0.204)	0.78 / 0.06	49.7 vs 51.3	
EM equities (EEM)	3,648	0.31	0.26	0.28	0.25	0.24	-1.2 (0.227)	0.88 / -0.10	46.5 vs 50.3	
20y+ Treasuries (TLT)	3,984	0.21	0.19	0.21	0.17	0.17	-1.7 (0.097)	0.79 / 0.18	28.4 vs 30.2	
gold (GLD)	3,984	0.35	0.26	0.26	0.22	0.21	-1.9 (0.057)	1.13 / -0.26	34.2 vs 37.6	
crude oil (USO)	3,984	0.39	0.31	0.34	0.25	0.24	-3.0 (0.003)	0.91 / 0.02	97.6 vs 101.0	
Apple	3,690	0.40	0.28	0.28	0.21	0.21	-3.4 (0.001)	0.88 / -0.05	62.7 vs 67.4	
Amazon	3,690	0.53	0.32	0.30	0.19	0.19	-7.4 (0.000)	0.95 / -0.07	61.7 vs 79.1	
Alphabet	3,690	0.50	0.31	0.29	0.23	0.22	-4.1 (0.000)	0.97 / -0.11	55.4 vs 63.7	
IBM	3,690	0.71	0.41	0.45	0.30	0.30	-4.1 (0.000)	1.03 / -0.13	65.9 vs 77.6	
Goldman Sachs	3,690	0.37	0.30	0.32	0.24	0.23	-2.1 (0.033)	1.05 / -0.35	64.4 vs 72.5	
Bitcoin	2,175	0.26	0.17	0.17	0.16	0.16	-1.1 (0.260)	0.77 / 0.09	100.6 vs 103.2	
Ether	1,963	0.26	0.20	0.19	0.21	0.20	0.9 (0.374)	0.67 / 0.07	142.8 vs 139.2	

Events

FOMC (62 decision days, 2019-2026): S&P |move| 0.95 % against 0.79 % on other days; RMS move over implied daily move (VIX / sqrt(252) the day before) 1.02 on FOMC days against 1.00 otherwise; TLT |move| 0.73 % vs 0.71 %, RMS over VXTLT-implied 0.88. VIX change on the day 0.05 pts (down on 56 % of days), VIX9D from the day before to the day after -0.84 pts. VIX1D (35 events since 2022): implied daily move 1.23 %, realised RMS 1.21 %, ratio 0.98.

Earnings (1,093 events): implied move from the front straddle by the 0.8 x straddle rule 5.38 %, stripped of the diffusion 6.01 %; realised |close-to-close| 4.65 % (RMS implied 6.77 vs RMS realised 6.37); implied above realised on 67 % of events, median ratio 1.44; Spearman(implied, |realised|) 0.43; realised^2 on implied^2 slope 0.46. Seller of the straddle: mean 0.27 % of spot per event, -0.43 % after the quoted spread, hit rate 55 %, sd 3.11 %, worst -25.52 %.

							year
2020	135	5.29	3.83	6.04 / 5.53	-0.51	60 %	
2021	146	4.13	3.44	4.66 / 4.50	-0.50	53 %	
2022	141	6.23	4.78	6.96 / 6.41	-0.13	57 %	
2023	167	5.77	4.49	6.68 / 5.84	0.03	59 %	
2024	178	6.54	5.42	7.29 / 7.46	-0.53	54 %	
2025	186	6.82	4.55	7.44 / 6.27	-0.19	56 %	
2026	140	6.99	5.89	7.55 / 7.80	-1.32	42 %	

Projectl - volatility dynamics research and systematic volatility strategies

name	n	implied (stripped) %	realised %	seller after spread %	hit
ABBV	27	4.26	3.56	-0.86	22 %
ADBE	27	8.30	6.32	-0.73	52 %
AMD	27	9.32	6.78	0.30	56 %
CSCO	27	6.94	5.58	-1.13	37 %
HD	27	4.23	2.67	0.22	74 %
ORCL	27	9.55	7.48	-0.88	44 %
PYPL	27	9.82	8.07	-0.24	59 %
QCOM	27	7.56	6.68	-1.14	41 %
SBUX	27	7.31	4.55	0.19	63 %
XOM	27	3.89	2.05	-0.38	67 %
BA	26	5.03	3.70	0.21	69 %
BAC	26	3.91	2.75	0.16	65 %
CAT	26	5.80	3.93	-0.31	58 %
CRM	26	8.45	7.00	-1.19	62 %
CVX	26	2.79	2.74	-0.72	46 %
INTC	26	9.02	9.45	-1.52	58 %
JNJ	26	2.88	1.98	-0.21	58 %
JPM	26	3.72	2.64	0.03	58 %
LLY	26	5.61	5.16	-1.29	38 %
MA	26	3.79	2.53	-0.16	58 %
MCD	26	3.53	1.82	0.27	65 %
MRK	26	4.05	2.92	-0.28	58 %
MSFT	26	5.45	4.45	-0.19	58 %
MU	26	9.34	6.50	0.24	65 %
NKE	26	8.60	8.12	-1.73	54 %
PFE	26	5.21	2.35	1.17	81 %
TXN	26	6.43	4.90	-0.63	46 %
UNH	26	3.90	5.04	-2.08	31 %
V	26	4.17	2.43	0.10	73 %
VZ	26	3.74	3.40	-0.90	50 %
DIS	25	7.10	5.50	-0.71	44 %
IBM	25	6.45	5.23	-1.34	40 %
KO	25	2.81	1.82	0.18	76 %
PEP	25	2.82	2.36	-0.43	36 %
AAPL	23	5.66	2.88	0.35	70 %
GS	23	3.57	2.14	0.10	65 %
COST	19	4.06	2.27	0.41	63 %
T	18	5.30	4.25	-0.83	44 %
AMZN	17	8.89	6.60	0.04	53 %
GOOGL	17	6.95	5.30	0.07	53 %
META	17	10.54	9.98	-2.09	47 %
TSLA	16	9.30	9.16	-1.40	44 %
GE	11	8.81	5.69	0.82	36 %
UBER	11	9.42	5.54	1.61	82 %
WMT	10	6.22	5.27	-0.61	40 %
AVGO	9	9.12	9.81	-3.36	11 %
NVDA	9	8.19	4.33	1.18	89 %

BTC calendar: weekend/weekday variance ratio 0.58 (annualised 55 % vs 72 %), by year 2018: 0.66, 2019: 0.82, 2020: 0.52, 2021: 0.69, 2022: 0.50, 2023: 0.41, 2024: 0.38, 2025: 0.39, 2026: 0.50; the half-hours after the 00/08/16 UTC funding marks run 1.28 x the variance of other hours. Implied: the 7-day vol on Fridays is -3.27 +/- 0.45 pts against the following Monday (lower on 69 % of 485 weeks).

ETH calendar: weekend/weekday variance ratio 0.61 (annualised 70 % vs 90 %), by year 2019: 0.68, 2020: 0.70, 2021: 0.61, 2022: 0.59, 2023: 0.51, 2024: 0.45, 2025: 0.55, 2026: 0.59; the half-hours after the 00/08/16 UTC funding marks run 1.41 x the variance of other hours. Implied: the 7-day vol on Fridays is -3.53 +/- 0.52 pts against the following Monday (lower on 69 % of 391 weeks).

The VIX complex

state (VIX9D / VIX / VIX3M)	share of days	mean VIX	realised next 21 days	premium (pts)	VIX change over 21 days
-----------------------------	---------------	----------	-----------------------	---------------	-------------------------

ProjectI - volatility dynamics research and systematic volatility strategies

backwardation	6.9 %	30.7	27.3	3.42	-5.91
contango	72.7 %	16.5	12.9	3.67	0.87
front kink (9D > VIX < 3M)	19.7 %	19.4	16.2	3.16	-0.87
hump (9D < VIX > 3M)	0.7 %	30.0	21.0	8.99	-7.65

VX curve 2013-05-20 to 2026-09-17, 3,356 days, in contango (F2 > F1) 85 % of days. Roll-down in index points per 30 days, mean (median): VIX->F1 1.15 (1.78), F1 0.89 (1.02), F2 0.51 (0.64), F3 0.35 (0.43), F4 0.28 (0.32).

Short front-month roll, one contract, 13.33 years: gross 12.1 pts/yr, bid-ask 0.54 pts/yr, net 11.6 pts/yr (\$11,584 a contract-year), Sharpe 0.58, max drawdown 64.6 pts, worst day -19.2 on 2020-03-16. Net per day when yesterday's curve was in contango 0.034 against 0.111 in backwardation; shorting only after contango days: 7.2 pts/yr, Sharpe 0.54.

														year
net pts	4.7	1.4	6.1	18.6	17.6	-16.8	31.0	-3.0	33.9	6.6	30.6	4.3	9.0	10.3

VVIX: mean 95.5; realised vol of the 30-day constant-maturity future over the next 21 days 64.0 (vol-of-vol premium 33.1 pts, CI [29.0, 37.0]); of the spot index 114.6. Correlation with the VIX level 0.64, with the 3M-1M slope -0.35; VVIX by slope tercile: backwardated 101.0, flat 90.0, steep 95.5.

Strategies

All programs: one unit (one straddle on one share / one coin), daily delta hedge, spread paid at entry (and exit where closed early), scenario margin = worst loss over +/-15 % spot x +/-30 % vol. Return on margin is net P&L per year over the average margin.

												program
SPY short straddle,	697	7.61	61 %	0.11	0.06	0.05	-10.5	0.62 / 0.38	4.0	68	0.24	
SPY straddle, VIX3M	643	7.61	61 %	0.16	0.05	0.10	-3.7	1.26 / -0.26	6.5	61	0.42	
SPY straddle, VIX3M	54	6.28	59 %	-0.47	0.12	-0.59	-10.5	-4.17 / 3.17	-7.0	86	-0.83	
SPY straddle, none	349	2.97	66 %	0.20	0.07	0.13	-10.5	0.81 / 0.19	8.9	44	0.67	
SPY straddle, none	348	4.71	55 %	0.02	0.05	-0.03	-5.1	0.17 / 0.83	0.7	56	0.05	
SPY straddle, stop,	0349	2.97	66 %	0.25	0.08	0.17	-7.3	1.72 / -0.72	11.5	35	0.88	
SPY straddle, stop,	0348	4.71	55 %	0.04	0.05	-0.01	-4.4	0.10 / 0.90	2.1	47	0.16	
SPY straddle, stop,	1349	2.97	66 %	0.20	0.07	0.13	-11.3	0.81 / 0.19	8.9	44	0.67	
SPY straddle, stop,	1348	4.71	55 %	0.02	0.05	-0.03	-4.7	0.00 / 1.00	0.8	55	0.06	
SPY straddle, stop,	2349	2.97	66 %	0.20	0.07	0.13	-10.5	0.81 / 0.19	8.9	44	0.67	
SPY straddle, stop,	2348	4.71	55 %	0.02	0.05	-0.03	-5.1	0.17 / 0.83	0.7	56	0.05	
SPY calendar (short	667	7.61	47 %	0.01	0.08	-0.07	-3.4	-10.71 / 11.71	-4.0	48	-0.49	
BTC short 30-day str	95	8.06	62 %	1.61	0.54	1.07	-15.1	0.78 / 0.22	49.4	121	0.74	
BTC straddle, iv60 >	71	8.06	59 %	1.20	0.54	0.66	-15.1	1.14 / -0.14	38.1	81	0.81	
BTC straddle, iv60 <	24	7.72	71 %	2.82	0.53	2.28	-8.7	-0.65 / 1.65	11.5	103	0.54	
BTC straddle, costle	95	8.06	64 %	1.61	0.20	1.41	-14.8	0.78 / 0.22	69.3	118	1.04	
ETH short 30-day str	88	7.47	51 %	0.18	0.54	-0.37	-23.6	0.57 / 0.43	-13.9	277	-0.12	
ETH straddle, iv60 >	64	7.04	55 %	0.49	0.54	-0.05	-14.5	-0.85 / 1.85	17.6	225	0.23	
ETH straddle, iv60 <	24	7.3	42 %	-0.65	0.55	-1.20	-23.6	1.95 / -2.95	-35.3	375	-0.85	
ETH straddle, costle	88	7.47	53 %	0.18	0.20	-0.03	-23.3	0.57 / 0.43	6.1	260	0.06	

Stop-loss chosen in sample: stop_0.5x_premium.

spy_straddle net by entry year: 2019: 34.4 (47, hit 74 %), 2020: 29.6 (153, hit 59 %), 2021: 108.2 (150, hit 69 %), 2022: -211.9 (139, hit 36 %), 2023: 77.9 (51, hit 78 %), 2024: -51.0 (58, hit 52 %), 2025: 81.1 (58, hit 74 %), 2026: 126.5 (41, hit 71 %)

btc_straddle net by entry year: 2018: 1138.0 (5, hit 60 %), 2019: 1613.6 (12, hit 50 %), 2020: 489.3 (12, hit 50 %), 2021: 10869.9 (11, hit 82 %), 2022: 844.7 (12, hit 67 %), 2023: -1933.9 (12, hit 58 %), 2024: 17901.3 (12, hit 75 %), 2025: 5376.0 (11, hit 55 %), 2026: -2558.1 (8, hit 62 %)